ONLINE MEDICAL STORE MANAGEMENT SYSTEM

Complete Project Documentation

Project Name: Online Medical Store Management System
Version: 1.0.0
Date: November 2025
Status: Development Complete

TABLE OF CONTENTS

---

1. INTRODUCTION OF DOCUMENT

1.1 Purpose and Overview

This document provides comprehensive technical and functional specifications for the **Online Medical Store Management System** - a full-stack web application designed to streamline the operations of an online pharmacy/medical store. The system integrates user management, medicine inventory, prescription verification, order processing, and multiple payment methods into a single cohesive platform.

1.2 Document Scope

This Software Requirements Specification (SRS) document serves as the primary reference for:

- **Development Team**: Implementation guidelines and technical architecture
- **Project Managers**: Scope, timeline, and resource planning
- **Quality Assurance**: Testing criteria and acceptance standards
- **Stakeholders**: System capabilities and deliverables

1.3 Benefits of This Phase

For Development:

- Clear technical specifications prevent misunderstandings during implementation
- Well-defined requirements reduce rework and development cycles
- Documentation serves as knowledge base for team members
- Standardized approach ensures consistency across modules

For Business:

- Clear scope prevents scope creep and cost overruns
- Documented requirements ensure stakeholder alignment
- Timeline visibility enables resource planning
- Risk identification and mitigation strategies

For Quality Assurance:

- Defined requirements enable comprehensive test case creation
- Clear acceptance criteria facilitate validation testing
- Documentation supports regression testing in later phases
- Traceability between requirements and test cases

1.4 Project Background

The Online Medical Store Management System addresses the growing demand for convenient access to medicines and medical products through digital channels. With increasing adoption of online healthcare services, this system provides:

- **Accessibility**: 24/7 availability of medicines and medical products
- **Convenience**: Home delivery of medications with prescription verification
- **Safety**: Pharmacist verification of prescriptions before fulfillment
- **Efficiency**: Automated inventory management and order processing
- **Compliance**: Regulatory adherence for prescription-based medicines

1.5 Key Stakeholders

Stakeholder	Role	Interests
End Users (Patients)	Customers	Easy ordering, prescription management, affordable prices
Pharmacists	Reviewers	Prescription verification, inventory management
Admin	System Manager	Dashboard analytics, user management, operations
Suppliers	Partners	Inventory updates, order fulfillment
Payment Providers	Service Providers	Transaction processing, security

---

2. SCOPE OF PROJECT

2.1 Project Scope Definition

The Online Medical Store Management System encompasses the development of a complete e-commerce platform specifically tailored for medical products and prescription medications with integrated pharmacy operations management.

2.2 In Scope

2.2.1 User Management Module

- User registration with email verification
- Secure login/logout functionality
- Profile management and address book
- Role-based access control (Patient, Pharmacist, Admin)
- Password reset and recovery mechanisms
- User activity tracking and audit logs

2.2.2 Medicine Catalog Module

- Comprehensive medicine database with 1000+ products
- Medicines categorization (pain relief, antibiotics, vitamins, etc.)
- Search and advanced filtering capabilities
- Medicine details (dosage, side effects, interactions)
- Stock level management
- Expiry date tracking and alerts

2.2.3 Prescription Management

- Prescription upload (PDF, JPG, PNG)
- Pharmacist verification workflow
- Prescription-based medicine restrictions
- Prescription history tracking
- Expiry management for prescriptions

2.2.4 Shopping Cart and Checkout

- Add/remove items from cart
- Quantity management
- Real-time price calculations
- Shipping address collection
- Order priority selection
- Medicine interaction warnings

2.2.5 Payment Processing

- **4 Payment Methods**:
  - Cash on Delivery (COD)
  - Credit/Debit Card (Stripe integration)
  - Bank Transfer
  - Mobile Wallet (JazzCash, EasyPaisa)
- Payment verification and status tracking
- Refund processing
- Transaction logging and audit trails

2.2.6 Order Management

- Order creation and confirmation
- Order tracking with timeline
- Shipping information management
- Order history and details
- Order cancellation policies
- Return and refund management

2.2.7 Admin Dashboard

- Dashboard with KPI metrics
- Medicine inventory management
- Order management and fulfillment
- User management and role assignment
- Prescription verification interface
- Supplier management
- 8 comprehensive report types

2.2.8 Reporting and Analytics

- **Sales Reports**: Revenue analysis by date range
- **Inventory Reports**: Stock levels and valuations
- **Expiry Reports**: Medicines expiring soon
- **Prescription Reports**: Verification statistics
- **Payment Reports**: Payment method analytics
- **User Activity Reports**: Spending patterns
- **Order Reports**: Order status distribution
- **Custom Filters**: Advanced filtering options

2.2.9 Notification System

- Order confirmation emails
- Prescription verification emails
- Password reset emails
- Order shipment notifications
- Welcome emails for new users
- SMS notifications (future phase)

2.2.10 Security Features

- JWT-based authentication (15-minute access tokens, 7-day refresh)
- Bcrypt password hashing
- Rate limiting (9 specialized limiters)
- CORS protection
- Helmet security headers
- Input validation and sanitization
- Role-based access control
- Audit logging

2.3 Out of Scope

Items Explicitly Excluded:

1. **Physical inventory warehouse management** - Supplier responsibility
2. **Logistics and delivery tracking** - Third-party integration (Phase 2)
3. **Video consultations** - Scheduled for future phase
4. **AI-powered prescription OCR** - Phase 2 enhancement
5. **Mobile app development** - Phase 2 development
6. **Telemedicine features** - Planned for Phase 3
7. **Insurance processing** - Future integration
8. **Government compliance automation** - Manual process initially

2.4 System Boundaries

Hardware Boundaries:

- Servers: Cloud-based (AWS/DigitalOcean)
- Client: Any modern web browser
- Storage: Cloud database (MySQL)
- File Storage: Cloud storage for prescriptions

Software Boundaries:

- Frontend: React 18.2 (Single Page Application)
- Backend: Node.js + Express API
- Database: MySQL 8.0
- Payment Gateway: Stripe API
- Email Service: SMTP (Nodemailer)

User Boundaries:

- **Maximum concurrent users**: 1000
- **Maximum medicines in database**: 5000
- **Maximum prescription file size**: 5MB

2.5 Constraints and Assumptions

Constraints:

1. **Budget Constraint**: Limited to initial feature set
2. **Time Constraint**: 4-month development timeline
3. **Technology Constraint**: Must use specified tech stack
4. **Compliance Constraint**: Must comply with healthcare data protection regulations
5. **Performance Constraint**: API response time < 200ms

Assumptions:

1. Internet connectivity available for all users
2. Users have valid email addresses for communication
3. Payment gateways remain stable and available
4. Regulatory environment remains unchanged
5. User data volume grows gradually
6. Suppliers provide timely inventory updates

2.6 Project Goals and Objectives

Primary Goals:

1. Launch a fully functional online medical store within 4 months
2. Support 1000+ medicines in initial database
3. Process 100+ orders daily by end of Year 1
4. Achieve 99.5% system uptime
5. Maintain customer satisfaction rating > 4.5/5

Specific Objectives:

1. Reduce manual order processing time from 30 minutes to 5 minutes

2. Implement automated prescription verification reducing processing time by 60%
3. Support 4 payment methods for maximum user convenience
4. Achieve sub-second page load times
5. Maintain comprehensive audit logs for compliance

---

3. FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

3.1 FUNCTIONAL REQUIREMENTS

FR1: User Authentication and Authorization

Requirement ID	FR1.1
Title	User Registration
Description	System shall allow users to register with email, password, name, and phone
Actor	User/Patient
Precondition	User is not registered
Main Flow	1. User enters details 2. System validates input 3. System creates user account 4. System sends verification email 5. User verifies email
Postcondition	User account created and verified
Priority	High
Acceptance Criteria	- Email must be unique - Password must meet security criteria - Verification email received - User account accessible

Requirement ID	FR1.2
Title	User Login
Description	System shall authenticate user credentials and generate JWT tokens
Actor	User/Pharmacist/Admin
Precondition	User account exists and verified
Main Flow	1. User enters email/password 2. System validates credentials 3. System generates access token 4. System generates refresh token 5. System stores tokens in frontend
Postcondition	User authenticated, tokens issued, user redirected to dashboard
Priority	High
Acceptance Criteria	- Login successful with correct credentials - Access token valid for 15 minutes - Refresh token valid for 7 days - Failed login shows error message

Requirement ID	FR1.3
Title	Role-Based Access Control
Description	System shall enforce different access levels based on user role
Actor	System
Precondition	User authenticated
Main Flow	1. User performs action 2. System checks user role 3. System verifies permissions 4. System grants/denies access
Postcondition	User can only access resources allowed by role
Priority	High
Acceptance Criteria	- Admin can access all features - Pharmacist can verify prescriptions - Patient can only view own orders - Unauthorized access blocked

FR2: Medicine Catalog Management

Requirement ID	FR2.1
Title	Browse Medicines
Description	System shall display medicines with search, filter, and sort capabilities
Actor	Patient
Precondition	User logged in
Main Flow	1. User navigates to medicines page 2. System displays all medicines 3. User applies filters/search 4. System returns filtered results 5. User views details
Postcondition	Filtered medicine list displayed
Priority	High
Acceptance Criteria	- Search works by name/description - Filter by category works - Sort by price/name works - Details page shows all information

Requirement ID	FR2.2
Title	Check Medicine Interactions
Description	System shall warn about potential drug interactions
Actor	Patient/System
Precondition	Multiple medicines in cart
Main Flow	1. User adds medicine to cart 2. System checks interactions with existing items 3. If interaction found, system shows warning 4. User acknowledges warning
Postcondition	User informed of interactions, can proceed with caution
Priority	High
Acceptance Criteria	- Interactions database populated - Warning shows severity level - User can override warning - Interaction logged

FR3: Prescription Management

Requirement ID	FR3.1
Title	Upload Prescription
Description	System shall allow users to upload prescription documents
Actor	Patient
Precondition	User logged in, has prescription file
Main Flow	1. User navigates to upload page 2. User selects file (PDF/JPG/PNG) 3. System validates file (< 5MB, correct format) 4. System stores file 5. System sends to pharmacist queue

Postcondition	Prescription uploaded, pending verification
Priority	High
Acceptance Criteria	- Only PDF/JPG/PNG accepted - File size limit 5MB enforced - Upload progress shown - Confirmation email sent

Requirement ID	FR3.2
Title	Pharmacist Verification
Description	System shall allow pharmacists to verify prescriptions
Actor	Pharmacist
Precondition	Prescription uploaded by patient
Main Flow	1. Pharmacist views pending prescriptions 2. Pharmacist reviews document 3. Pharmacist approves/rejects 4. System sends notification to patient 5. Status updated in database
Postcondition	Prescription status updated, patient notified
Priority	High
Acceptance Criteria	- Only pharmacist can verify - Verification date recorded - Rejection reason captured - Patient notified within 1 hour

FR4: Shopping Cart and Checkout

Requirement ID	FR4.1
Title	Add/Remove Items from Cart
Description	System shall manage shopping cart items
Actor	Patient
Precondition	User logged in, viewing medicine
Main Flow	1. User clicks add to cart 2. System adds item to cart 3. Cart updated in real-time 4. User can adjust quantity or remove
Postcondition	Cart reflects user selections
Priority	High
Acceptance Criteria	- Add item increments quantity - Remove item removes from cart - Quantity updates price automatically - Out of stock items handled

Requirement ID	FR4.2
Title	Checkout Process
Description	System shall process checkout with shipping and payment details
Actor	Patient
Precondition	Items in cart, user logged in
Main Flow	1. User clicks checkout 2. System collects shipping address 3. System selects payment method 4. System calculates total (subtotal + shipping + tax) 5. System confirms order
Postcondition	Order created, payment processed, confirmation sent
Priority	High
Acceptance Criteria	- All required fields validated - Shipping cost calculated - Tax calculated (5%) - Order confirmation email sent

FR5: Payment Processing

Requirement ID	FR5.1
Title	Multiple Payment Methods
Description	System shall support 4 payment methods
Actor	Patient/System
Precondition	User at checkout
Main Flow	1. User selects payment method 2. System displays appropriate payment form 3. User enters payment details 4. System processes payment 5. System confirms transaction
Postcondition	Payment processed, order confirmed
Priority	High
Acceptance Criteria	- COD option available - Card payment integrated with Stripe - Bank transfer details provided - Wallet payment options shown

FR6: Order Management

Requirement ID	FR6.1
Title	Order Tracking
Description	System shall provide order status tracking with timeline
Actor	Patient
Precondition	Order placed
Main Flow	1. User navigates to orders page 2. System displays order history 3. User selects order 4. System shows order details with timeline 5. User views current status
Postcondition	User sees complete order information
Priority	High
Acceptance Criteria	- Order list shows all orders - Status timeline visible - Tracking information available - Estimated delivery shown

FR7: Admin Dashboard

Requirement ID	FR7.1
Title	Dashboard Statistics
Description	System shall display key performance indicators on admin dashboard
Actor	Admin
Precondition	Admin logged in
Main Flow	1. Admin navigates to dashboard 2. System queries database for metrics 3. System calculates KPIs 4. System displays dashboard cards

Postcondition	Dashboard displayed with current metrics
Priority	High
Acceptance Criteria	- Total products count shown - Total orders count shown - Total revenue displayed - Total users count shown - Real-time updates

Requirement ID	**FR7.2**
Title	Generate Reports
Description	System shall generate 8 types of business reports
Actor	Admin
Precondition	Admin has data in system
Main Flow	1. Admin selects report type 2. Admin selects date range (optional) 3. System queries database 4. System generates report 5. System displays/downloads report
Postcondition	Report generated and available for download
Priority	High
Acceptance Criteria	- Sales report shows revenue trends - Inventory report shows stock levels - Expiry report shows products expiring soon - Prescription report shows verification rate - All reports filterable by date

FR8: Notification System

Requirement ID	**FR8.1**
Title	Email Notifications
Description	System shall send automated email notifications
Actor	System
Precondition	User action triggers notification event
Main Flow	1. User action triggers event 2. System queues notification 3. System generates email 4. System sends via SMTP 5. System logs notification
Postcondition	Email sent to user
Priority	Medium
Acceptance Criteria	- Order confirmation email sent - Prescription verification email sent - Password reset email sent - Email contains all relevant details

3.2 NON-FUNCTIONAL REQUIREMENTS

NFR1: Performance Requirements

Requirement ID	**NFR1.1**
Title	API Response Time
Description	API endpoints shall respond within 200 milliseconds for 95% of requests
Metric	Response time in milliseconds
Target	< 200ms for 95th percentile
Measurement	Application Performance Monitoring (APM) tools
Priority	High

Requirement ID	**NFR1.2**
Title	Page Load Time
Description	Frontend pages shall load in under 3 seconds
Metric	Time to Interactive (TTI)
Target	< 3 seconds for 90% of users
Measurement	Google Lighthouse, WebPageTest
Priority	High

Requirement ID	**NFR1.3**
Title	Database Query Performance
Description	Database queries shall execute within 100ms
Metric	Query execution time
Target	< 100ms for 95% of queries
Measurement	Database monitoring tools
Priority	High

NFR2: Scalability Requirements

Requirement ID	**NFR2.1**
Title	Concurrent User Support
Description	System shall support 1000 concurrent users without degradation
Metric	Number of concurrent connections
Target	Minimum 1000 concurrent users
Measurement	Load testing tools
Priority	High

Requirement ID	**NFR2.2**
Title	Data Volume Capacity
Description	System shall handle 5000 medicines and 100,000 orders
Metric	Database records
Target	5000 medicines, 100,000 orders
Measurement	Database monitoring
Priority	Medium

NFR3: Availability and Reliability

Requirement ID	**NFR3.1**
Title	System Uptime

Description	System shall maintain 99.5% uptime excluding planned maintenance
Metric	Uptime percentage
Target	99.5% (approximately 3.6 hours downtime/month)
Measurement	Uptime monitoring services
Priority	High

Requirement ID	**NFR3.2**
Title	Disaster Recovery
Description	System shall recover from failures within 1 hour
Metric	Recovery Time Objective (RTO)
Target	RTO < 1 hour
Measurement	Disaster recovery testing
Priority	High

Requirement ID	**NFR3.3**
Title	Data Backup
Description	System shall backup data daily with retention of 30 days
Metric	Backup frequency and retention
Target	Daily backups, 30-day retention
Measurement	Backup logs and monitoring
Priority	High

NFR4: Security Requirements

Requirement ID	**NFR4.1**
Title	Authentication Security
Description	Passwords shall be hashed using bcrypt with minimum 10 salt rounds
Metric	Hashing algorithm and iterations
Target	bcrypt with 10+ salt rounds
Measurement	Code review, security testing
Priority	Critical

Requirement ID	**NFR4.2**
Title	Token Security
Description	JWT tokens shall have 15-minute expiration for access tokens
Metric	Token expiration time
Target	Access token: 15 minutes, Refresh token: 7 days
Measurement	Code review, token analysis
Priority	Critical

Requirement ID	**NFR4.3**
Title	Rate Limiting
Description	System shall implement rate limiting to prevent brute force attacks
Metric	Requests per time window
Target	Login: 5 attempts/15min, Register: 3/hour, General: 100/15min
Measurement	Rate limiter configuration review
Priority	High

Requirement ID	**NFR4.4**
Title	Data Encryption
Description	Sensitive data (passwords, tokens) shall be encrypted in transit (HTTPS)
Metric	Encryption protocol
Target	TLS 1.2 or higher for all connections
Measurement	SSL/TLS certificate verification
Priority	Critical

NFR5: Usability Requirements

Requirement ID	**NFR5.1**
Title	Responsive Design
Description	System shall be responsive and work on mobile, tablet, and desktop
Metric	Device compatibility
Target	Support iOS 12+, Android 6+, modern browsers
Measurement	Cross-device testing
Priority	High

Requirement ID	**NFR5.2**
Title	Accessibility Compliance
Description	System shall comply with WCAG 2.1 AA accessibility standards
Metric	Accessibility score
Target	WCAG 2.1 Level AA compliance
Measurement	Automated accessibility testing, manual review
Priority	Medium

Requirement ID	**NFR5.3**
Title	User Interface Consistency
Description	UI shall follow consistent design patterns throughout application
Metric	Design consistency score
Target	100% UI consistency

Measurement	Design review, user testing
Priority	Medium

NFR6: Maintainability Requirements

Requirement ID	**NFR6.1**
Title	Code Documentation
Description	Code shall be well-documented with comments and API documentation
Metric	Documentation coverage
Target	80% of functions documented
Measurement	Documentation review
Priority	Medium

Requirement ID	**NFR6.2**
Title	Code Quality
Description	Code shall follow linting standards with < 5 warnings per module
Metric	Linting warnings count
Target	< 5 warnings per module
Measurement	ESLint analysis
Priority	Medium

NFR7: Compliance Requirements

Requirement ID	**NFR7.1**
Title	Data Protection Compliance
Description	System shall comply with data protection regulations
Metric	Compliance level
Target	GDPR-compliant, user data protection
Measurement	Security audit, legal review
Priority	Critical

Requirement ID	**NFR7.2**
Title	Healthcare Regulations
Description	System shall comply with healthcare data protection standards
Metric	Compliance level
Target	Prescription-based access control, audit trails
Measurement	Healthcare compliance audit
Priority	Critical

---

4. USE CASE DIAGRAM

4.1 System Actors

Actor	Role	Description
Patient/Customer	Primary Actor	End user purchasing medicines and managing prescriptions
Pharmacist	Secondary Actor	Verifies prescriptions before order fulfillment
Admin/Manager	Secondary Actor	Manages system, inventory, users, and generates reports
Supplier	Secondary Actor	Manages inventory and product information
Payment Provider	External System	Processes credit card and payment transactions
Email Service	External System	Sends notifications and confirmations

4.2 Use Cases Overview

Patient Use Cases:

1. **UC1: Register Account** - Create new user account
2. **UC2: Login** - Access system with credentials
3. **UC3: Browse Medicines** - Search and view medicine catalog
4. **UC4: Check Interactions** - Check medicine interactions
5. **UC5: Upload Prescription** - Upload prescription file
6. **UC6: Add to Cart** - Add medicines to shopping cart
7. **UC7: Checkout** - Complete purchase
8. **UC8: Select Payment Method** - Choose payment option
9. **UC9: Track Order** - View order status and tracking
10. **UC10: View Profile** - Manage personal information

Pharmacist Use Cases:

1. **UC11: Verify Prescription** - Review and verify uploaded prescriptions
2. **UC12: View Orders** - Monitor orders requiring prescriptions
3. **UC13: Manage Inventory** - Update stock levels

Admin Use Cases:

1. **UC14: View Dashboard** - See system statistics
2. **UC15: Manage Medicines** - Add/edit/delete medicines
3. **UC16: Manage Users** - View and manage user accounts
4. **UC17: Manage Orders** - Monitor and manage orders
5. **UC18: Generate Reports** - Create business reports

6. **UC19: View Analytics** - Monitor system performance

4.3 Use Case Descriptions

UC1: Register Account

```
Actor: Patient
Preconditions: User not registered, access to registration page
Main Flow:
1. User enters name, email, phone, password
2. System validates input (unique email, strong password)
3. System creates user account
4. System sends verification email
5. User verifies email by clicking link
6. Account activated
Postconditions: User account created and verified
Alternate Flow: Email already registered → Show error message
```

UC2: Login

```
Actor: Patient/Pharmacist/Admin
Preconditions: User account exists and verified
Main Flow:
1. User enters email and password
2. System authenticates credentials
3. System generates JWT tokens
4. User redirected to dashboard
Postconditions: User authenticated, tokens stored
Alternate Flow: Wrong credentials → Show error, allow retry
```

UC3: Browse Medicines

```
Actor: Patient
Preconditions: User logged in or browsing as guest
Main Flow:
1. User navigates to medicines page
2. System displays medicine list (paginated)
3. User applies filters (category, price)
4. User performs search
5. System displays filtered results
6. User clicks medicine for details
Postconditions: Medicine details displayed
```

UC8: Select Payment Method

```
Actor: Patient
Preconditions: Items in cart, at checkout page
Main Flow:
1. User selects payment method from options
2. If COD: Order placed directly
3. If Card: System shows Stripe form
4. If Bank Transfer: System shows bank details
5. If Wallet: System shows wallet options
6. User enters/confirms payment
7. System processes payment
Postconditions: Payment processed, order created
```

UC11: Verify Prescription

```
Actor: Pharmacist
Preconditions: Prescription uploaded by patient
Main Flow:
1. Pharmacist views pending prescriptions queue
2. Pharmacist reviews prescription document
3. Pharmacist checks prescription validity (date, doctor)
4. Pharmacist approves or rejects
5. If approved: Mark as verified, notify patient
6. If rejected: Record reason, notify patient
Postconditions: Prescription status updated
```

UC14: View Dashboard

```
Actor: Admin
Preconditions: Admin logged in
Main Flow:
1. Admin navigates to dashboard
2. System queries current metrics
3. System displays KPI cards:
   - Total products
   - Total orders
   - Total revenue
   - Total users
4. System displays charts and trends
Postconditions: Dashboard displayed
```

UC18: Generate Reports

```
Actor: Admin
Preconditions: Admin logged in, has data
Main Flow:
1. Admin selects report type from menu
2. Admin specifies date range (optional)
3. System queries database
4. System calculates metrics
5. System displays/exports report
Postconditions: Report available for download
Report Types: Sales, Inventory, Expiry, Prescription, Payment, User Activity, Order, Custom
```

4.4 Use Case Diagram (Text Representation)

```
                            ┌──────────────────────────────────────┐
                            │     Online Medical Store System       │
                            └──────────────────────────────────────┘

┌──────────────┐                                                ┌──────────────┐
│   Patient    │                                                │  Pharmacist  │
└──────────────┘                                                └──────────────┘
       │                                                               │
       │                                                                │
       ├──────UC1: Register Account                                    │
       ├──────UC2: Login                                               │
       ├──────UC3: Browse Medicines                                    ├─UC11: Verify Prescription
       ├──────UC4: Check Interactions                                  ├─UC12: View Orders
       ├──────UC5: Upload Prescription                                 ├─UC13: Manage Inventory
       ├──────UC6: Add to Cart                              ┌──────────────┐
       ├──────UC7: Checkout                                 │    Admin     │
       ├──────UC8: Select Payment Method                    └──────────────┘
       ├──────UC9: Track Order                                     │
       └──────UC10: View Profile                                   ├─UC14: View Dashboard
                                                                   ├─UC15: Manage Medicines
                                                                   ├─UC16: Manage Users
                                                                   ├─UC17: Manage Orders
                                                                   ├─UC18: Generate Reports
                                                                   └─UC19: View Analytics

                            External Systems:
                    ┌──────────────────────────────┐
                    │   Payment Provider (Stripe)  │
                    │   Email Service (Nodemailer)│
                    │   SMS Service (Twilio)       │
                    └──────────────────────────────┘
```

---

5. WORK PLAN (GANTT CHART)

5.1 Project Timeline

Project Duration: 16 weeks (4 months)
Start Date: Week 1
End Date: Week 16

5.2 Project Phases

Phase 1: Planning & Analysis (Weeks 1-2)

- Requirement gathering
- System design
- Technology selection
- Team allocation
- Infrastructure planning

Phase 2: Development - Backend (Weeks 3-7)

- Database design and setup
- API development
- Authentication and authorization
- Payment integration
- Email service setup

Phase 3: Development - Frontend (Weeks 5-10)

- UI/UX design
- Component development
- Page implementation
- Form validation
- State management

Phase 4: Integration & Testing (Weeks 11-13)

- Integration testing
- End-to-end testing
- Security testing
- Performance testing
- User acceptance testing

Phase 5: Deployment & Launch (Weeks 14-16)

- Production deployment
- User training
- Documentation
- Monitoring setup
- Go-live support

5.3 Gantt Chart (Text Representation)

```
Project: Online Medical Store Management System
Timeline: 16 Weeks

WEEK:           1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16

Phase 1: Planning & Analysis
├─ Requirement Gathering    [████░░░░░░░░░░░░░░░░░░░░░░░░░░]
├─ System Design            [░████░░░░░░░░░░░░░░░░░░░░░░░░░]
├─ Technology Selection     [░░████░░░░░░░░░░░░░░░░░░░░░░░░]
└─ Infrastructure Setup     [░░░░████░░░░░░░░░░░░░░░░░░░░░░]

Phase 2: Backend Development
├─ Database Design          [░░░████░░░░░░░░░░░░░░░░░░░░░░░]
├─ API Development          [░░░░████████░░░░░░░░░░░░░░░░░░]
├─ Authentication          [░░░░░████████░░░░░░░░░░░░░░░░░]
├─ Payment Integration     [░░░░░░░░████░░░░░░░░░░░░░░░░░░]
└─ Email Service           [░░░░░░░░░████░░░░░░░░░░░░░░░░░]

Phase 3: Frontend Development
├─ UI/UX Design            [░░░░████░░░░░░░░░░░░░░░░░░░░░░]
├─ Component Development   [░░░░░░████████░░░░░░░░░░░░░░░░]
├─ Page Implementation     [░░░░░░░░████████░░░░░░░░░░░░░░]
├─ Form Validation         [░░░░░░░░░░░████░░░░░░░░░░░░░░░]
└─ State Management        [░░░░░░░░░░░████░░░░░░░░░░░░░░░]

Phase 4: Testing & Integration
├─ Integration Testing     [░░░░░░░░░░░░░████░░░░░░░░░░░░░]
├─ E2E Testing            [░░░░░░░░░░░░░░████░░░░░░░░░░░░░]
├─ Security Testing       [░░░░░░░░░░░░░░░████░░░░░░░░░░░░]
├─ Performance Testing    [░░░░░░░░░░░░░░░░░████░░░░░░░░░░]
└─ UAT                    [░░░░░░░░░░░░░░░░░░████░░░░░░░░░]

Phase 5: Deployment
├─ Staging Deployment     [░░░░░░░░░░░░░░░░░░░░████░░░░░░░]
├─ Production Deploy      [░░░░░░░░░░░░░░░░░░░░░░████░░░░░]
├─ User Training          [░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░]
└─ Go-Live Support        [░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░]
```

5.4 Milestone Schedule

Milestone	Week	Status	Deliverable
Requirements Complete	2	✓	SRS Document
Database Schema Ready	4	✓	ERD, Database Design
Backend APIs Ready	8	✓	API Documentation
Frontend Components Done	10	✓	Component Library
All Tests Pass	13	⏳	Test Reports
Production Ready	16	⏳	System Go-Live

5.5 Resource Allocation

Role	Team Size	Weeks
Project Manager	1	16
Backend Developer	2	16
Frontend Developer	2	16
QA Engineer	1	12
DevOps Engineer	1	8
UI/UX Designer	1	4

---

6. ENTITY RELATIONSHIP DIAGRAM (ERD)

6.1 Database Overview

The system uses 8 core tables with relationships to manage users, medicines, orders, prescriptions, and payments.

6.2 Entity Descriptions

USERS Table

```
USERS
├─ id (Primary Key)
├─ name (VARCHAR 100)
├─ email (VARCHAR 150, UNIQUE)
├─ phone (VARCHAR 20, UNIQUE, NULLABLE)
├─ password_hash (VARCHAR 255)
├─ role (ENUM: patient, doctor, pharmacist, admin)
├─ is_verified (TINYINT)
├─ created_at (TIMESTAMP)
└─ updated_at (TIMESTAMP)
```

MEDICINES Table

```
MEDICINES
├─ id (Primary Key)
├─ name (VARCHAR 150)
├─ description (TEXT)
├─ price (DECIMAL 10,2)
├─ stock (INT)
├─ requires_prescription (TINYINT)
├─ image_url (VARCHAR 255)
├─ category (VARCHAR 100)
├─ expiry_date (DATE, NULLABLE)
├─ supplier_id (Foreign Key → SUPPLIERS)
├─ dosage_instructions (TEXT)
├─ side_effects (TEXT)
├─ created_at (TIMESTAMP)
└─ updated_at (TIMESTAMP)
```

PRESCRIPTIONS Table

```
PRESCRIPTIONS
├─ id (Primary Key)
├─ user_id (Foreign Key → USERS)
├─ file_path (VARCHAR 255)
├─ file_original_name (VARCHAR 255)
├─ file_mime_type (VARCHAR 100)
├─ file_size (BIGINT)
├─ status (ENUM: pending, verified, rejected, expired)
├─ notes (TEXT, NULLABLE)
├─ uploaded_at (TIMESTAMP)
├─ verified_at (TIMESTAMP, NULLABLE)
└─ verified_by (BIGINT, NULLABLE → USERS)
```

ORDERS Table

```
ORDERS
├─ id (Primary Key)
├─ user_id (Foreign Key → USERS)
├─ total_amount (DECIMAL 12,2)
├─ status (ENUM: pending, confirmed, shipped, delivered, cancelled)
├─ shipping_address (TEXT)
├─ shipping_city (VARCHAR 100)
├─ shipping_postal_code (VARCHAR 20)
├─ payment_method (ENUM: cod, card, bank, wallet)
├─ prescription_id (Foreign Key → PRESCRIPTIONS, NULLABLE)
├─ order_date (TIMESTAMP)
├─ delivery_date (TIMESTAMP, NULLABLE)
├─ priority (ENUM: normal, high, urgent)
├─ created_at (TIMESTAMP)
└─ updated_at (TIMESTAMP)
```

ORDER_ITEMS Table

```
ORDER_ITEMS
├─ id (Primary Key)
├─ order_id (Foreign Key → ORDERS)
├─ medicine_id (Foreign Key → MEDICINES)
├─ quantity (INT)
├─ price_at_purchase (DECIMAL 10,2)
├─ created_at (TIMESTAMP)
└─ updated_at (TIMESTAMP)
```

PAYMENTS Table

```
PAYMENTS
├─ id (Primary Key)
├─ order_id (Foreign Key → ORDERS, UNIQUE)
├─ method (ENUM: cod, card, bank, wallet)
├─ amount (DECIMAL 12,2)
├─ status (ENUM: pending, completed, failed, refunded)
├─ transaction_id (VARCHAR 255, NULLABLE)
├─ stripe_payment_id (VARCHAR 255, NULLABLE)
├─ refund_amount (DECIMAL 12,2, NULLABLE)
├─ refund_reason (VARCHAR 255, NULLABLE)
├─ payment_date (TIMESTAMP)
├─ refund_date (TIMESTAMP, NULLABLE)
├─ created_at (TIMESTAMP)
└─ updated_at (TIMESTAMP)
```

INTERACTIONS Table

```
INTERACTIONS
├─ id (Primary Key)
├─ medicine1_id (Foreign Key → MEDICINES)
├─ medicine2_id (Foreign Key → MEDICINES)
├─ severity (ENUM: minor, moderate, severe)
├─ description (TEXT)
├─ created_at (TIMESTAMP)
└─ updated_at (TIMESTAMP)
```

SUPPLIERS Table

```
SUPPLIERS
├─ id (Primary Key)
├─ name (VARCHAR 150)
├─ contact_person (VARCHAR 100)
├─ email (VARCHAR 150)
├─ phone (VARCHAR 20)
├─ address (TEXT)
├─ city (VARCHAR 100)
├─ state (VARCHAR 100)
├─ postal_code (VARCHAR 20)
├─ created_at (TIMESTAMP)
└─ updated_at (TIMESTAMP)
```

PASSWORD_RESET_TOKENS Table

```
PASSWORD_RESET_TOKENS
├─ id (Primary Key)
├─ user_id (Foreign Key → USERS)
├─ token_hash (VARCHAR 255)
├─ expires_at (DATETIME)
├─ used (TINYINT)
└─ created_at (TIMESTAMP)
```

6.3 ERD (Text Representation)

```
                    ┌────────────────┐
                    │     USERS      │
                    ├────────────────┤
                    │ PK: id         │
                    │ name           │
                    │ email (UNIQUE) │
                    │ phone          │
                    │ password_hash  │
                    │ role           │
                    │ is_verified    │
                    │ created_at     │
                    │ updated_at     │
                    └────────────────┘
                            │
          ┌─────────────────┼─────────────────┐
          │                 │                 │
          ▼                 ▼                 │
┌─────────────────┐   ┌────────────────┐     │
│  PRESCRIPTIONS  │   │  ORDERS        │     │
├─────────────────┤   ├────────────────┤     │
│ PK: id          │   │ PK: id         │     │
│ FK: user_id     │   │ FK: user_id    │     │
│ file_path       │   │ FK: prescription_id  │
│ status          │   │ total_amount   │     │
│ verified_at     │   │ status         │     │
│ FK: verified_by │   │ payment_method │     │
└─────────────────┘   │ priority       │     │
                      │ delivery_date  │     │
                      └────────────────┘     │
                              │              │
              ┌───────────────┼───┐          │
              │               │   │          │
              ▼               ▼   ▼          │
      ┌────────────┐   ┌─────────────┐      │
      │ ORDER_ITEMS│   │ PAYMENTS    │      │
      ├────────────┤   ├─────────────┤      │
      │ PK: id     │   │ PK: id      │      │
      │ FK: order_id   │ FK: order_id│      │
      │ FK: medicine_id│ method      │      │
      │ quantity   │   │ amount      │      │
      │ price      │   │ status      │      │
      └────────────┘   │ transaction_id     │
              │        │ stripe_id   │      │
              │        └─────────────┘      │
              ▼                             │
      ┌────────────────┐                    │
      │   MEDICINES    │                    │
      ├────────────────┤                    │
      │ PK: id         │                    │
      │ name           │                    │
      │ description    │                    │
      │ price          │                    │
      │ stock          │                    │
      │ requires_prescription               │
      │ FK: supplier_id│                    │
      │ category       │                    │
      │ expiry_date    │                    │
      │ dosage_instructions                 │
      └────────────────┘                    │
              │                             │
              ├─────────────────────────────┘
              ▼
      ┌────────────────┐
      │  INTERACTIONS  │
      ├────────────────┤
      │ PK: id         │
      │ FK: medicine1_id
      │ FK: medicine2_id
      │ severity       │
      │ description    │
      └────────────────┘
              │
              ▼
      ┌────────────────┐
      │   SUPPLIERS    │
      ├────────────────┤
      │ PK: id         │
      │ name           │
      │ contact_person │
      │ email          │
      │ phone          │
      │ address        │
      │ city           │
      └────────────────┘
```

6.4 Relationships Summary

Relationship	From	To	Type	Cardinality
User → Prescriptions	USERS	PRESCRIPTIONS	1:N	

				One user many prescriptions
User → Orders	USERS	ORDERS	1:N	One user many orders
Order → OrderItems	ORDERS	ORDER_ITEMS	1:N	One order many items
Order → Payments	ORDERS	PAYMENTS	1:1	One order one payment
Medicine → OrderItems	MEDICINES	ORDER_ITEMS	1:N	One medicine many order items
Medicine → Interactions	MEDICINES	INTERACTIONS	N:M	Medicines can interact
Supplier → Medicines	SUPPLIERS	MEDICINES	1:N	One supplier many medicines
Prescription → Orders	PRESCRIPTIONS	ORDERS	1:N	One prescription many orders

---

7. DATABASE DESIGN

7.1 Database Schema Details

7.1.1 Data Types Used

```
BIGINT       - Large integers for IDs and counts (8 bytes)
INT          - Regular integers for quantities (4 bytes)
VARCHAR(n)   - Variable-length strings (max n characters)
TEXT         - Long text fields for descriptions
DECIMAL(p,s) - Fixed-point decimals (p=precision, s=scale)
TIMESTAMP    - Date and time with timezone
DATE         - Date only (no time)
DATETIME     - Date and time
TINYINT(1)   - Boolean values (0/1) or small integers
ENUM         - Enumerated values (predefined list)
```

7.1.2 Indexing Strategy

```
Indexed Columns:
├─ USERS.email (UNIQUE INDEX)
├─ USERS.phone (UNIQUE INDEX)
├─ USERS.role (REGULAR INDEX)
├─ MEDICINES.category (REGULAR INDEX)
├─ MEDICINES.expiry_date (REGULAR INDEX)
├─ ORDERS.user_id (REGULAR INDEX)
├─ ORDERS.status (REGULAR INDEX)
├─ ORDERS.created_at (REGULAR INDEX)
├─ ORDER_ITEMS.order_id (FOREIGN KEY INDEX)
├─ ORDER_ITEMS.medicine_id (FOREIGN KEY INDEX)
├─ PAYMENTS.order_id (FOREIGN KEY INDEX)
├─ PAYMENTS.status (REGULAR INDEX)
├─ PRESCRIPTIONS.user_id (FOREIGN KEY INDEX)
├─ PRESCRIPTIONS.status (REGULAR INDEX)
└─ SUPPLIERS.email (REGULAR INDEX)
```

7.1.3 Constraints and Validations

```
PRIMARY KEY Constraints:
- Each table has exactly one PRIMARY KEY (id)

UNIQUE Constraints:
- USERS.email UNIQUE (prevents duplicate emails)
- USERS.phone UNIQUE (prevents duplicate phones)
- PAYMENTS.order_id UNIQUE (one payment per order)

FOREIGN KEY Constraints:
- PRESCRIPTIONS.user_id → USERS.id (CASCADE DELETE)
- ORDERS.user_id → USERS.id (CASCADE DELETE)
- ORDERS.prescription_id → PRESCRIPTIONS.id (SET NULL)
- ORDER_ITEMS.order_id → ORDERS.id (CASCADE DELETE)
- ORDER_ITEMS.medicine_id → MEDICINES.id (SET NULL)
- PAYMENTS.order_id → ORDERS.id (CASCADE DELETE)
- MEDICINES.supplier_id → SUPPLIERS.id (SET NULL)
- INTERACTIONS.medicine1_id → MEDICINES.id (CASCADE DELETE)
- INTERACTIONS.medicine2_id → MEDICINES.id (CASCADE DELETE)

NOT NULL Constraints:
- USERS.name, email, password_hash, role
- MEDICINES.name, price, stock
- ORDERS.user_id, total_amount, status
- ORDER_ITEMS.order_id, medicine_id, quantity, price_at_purchase
- PAYMENTS.order_id, method, amount, status
- PRESCRIPTIONS.user_id, file_path, status

DEFAULT Values:
- USERS.role DEFAULT 'patient'
- USERS.is_verified DEFAULT 0
- MEDICINES.stock DEFAULT 0
- MEDICINES.requires_prescription DEFAULT 0
- ORDERS.status DEFAULT 'pending'
- ORDERS.priority DEFAULT 'normal'
- PAYMENTS.status DEFAULT 'pending'
- PRESCRIPTIONS.status DEFAULT 'pending'
- All timestamp columns DEFAULT CURRENT_TIMESTAMP
```

7.1.4 View Definitions

```sql
-- View: Medicine Stock Status
CREATE VIEW medicine_stock_status AS
SELECT id, name, stock, category,
  CASE
    WHEN stock = 0 THEN 'Out of Stock'
    WHEN stock < 10 THEN 'Low Stock'
    ELSE 'In Stock'
  END AS status
FROM medicines;

-- View: Pending Prescriptions
CREATE VIEW pending_prescriptions AS
SELECT p.id, u.name, u.email, p.uploaded_at
FROM prescriptions p
JOIN users u ON p.user_id = u.id
WHERE p.status = 'pending'
ORDER BY p.uploaded_at ASC;

-- View: Recent Orders
CREATE VIEW recent_orders AS
SELECT o.id, u.name, o.total_amount, o.status, o.created_at
FROM orders o
JOIN users u ON o.user_id = u.id
ORDER BY o.created_at DESC
LIMIT 100;

-- View: Monthly Revenue
CREATE VIEW monthly_revenue AS
SELECT
  DATE_TRUNC('month', created_at) AS month,
  SUM(total_amount) AS revenue,
  COUNT(*) AS order_count
FROM orders
WHERE status = 'delivered'
GROUP BY DATE_TRUNC('month', created_at);
```

7.2 Database Normalization

7.2.1 Normalization Level: 3NF (Third Normal Form)

First Normal Form (1NF) - Atomic Values:

- All columns contain atomic (indivisible) values
- No multi-valued attributes
- Example: phone column has single value, not list

Second Normal Form (2NF) - No Partial Dependencies:

- Meets 1NF requirements
- Every non-key attribute depends on entire primary key
- Example: Order items have both order_id and medicine_id

Third Normal Form (3NF) - No Transitive Dependencies:

- Meets 2NF requirements
- Non-key attributes don't depend on other non-key attributes
- Example: supplier_id in medicines, not supplier name

7.3 Database Capacity Planning

7.3.1 Storage Requirements

```
Estimated Row Counts:
├─ USERS: 10,000 rows (~500 KB)
├─ MEDICINES: 5,000 rows (~1 MB)
├─ ORDERS: 100,000 rows (~5 MB)
├─ ORDER_ITEMS: 300,000 rows (~9 MB)
├─ PRESCRIPTIONS: 50,000 rows (~10 MB, with file references)
├─ PAYMENTS: 100,000 rows (~4 MB)
├─ INTERACTIONS: 5,000 rows (~250 KB)
└─ SUPPLIERS: 500 rows (~50 KB)

Total Database Size: ~30 GB (with file uploads)
```

7.3.2 Backup and Recovery

```
Backup Strategy:
├─ Daily full backups (executed at 2 AM)
├─ Hourly incremental backups
├─ 30-day retention policy
├─ Offsite backup storage
├─ Monthly full backup archival
└─ Automated backup verification

Recovery Time Objectives (RTO):
├─ RTO for data loss: 1 hour
├─ RPO (Recovery Point Objective): 15 minutes
└─ Backup restoration time: < 30 minutes
```

7.4 SQL Queries for Key Operations

7.4.1 Sample Queries

```sql
-- Query 1: Get user order history with items
SELECT
  o.id,
  o.total_amount,
  o.status,
  o.created_at,
  mi.name,
  oi.quantity,
  oi.price_at_purchase
FROM orders o
JOIN order_items oi ON o.id = oi.order_id
JOIN medicines mi ON oi.medicine_id = mi.id
WHERE o.user_id = ?
ORDER BY o.created_at DESC;

-- Query 2: Find low stock medicines
SELECT
  id,
  name,
  stock,
  category
FROM medicines
WHERE stock < 20
ORDER BY stock ASC;

-- Query 3: Get expiring medicines within 30 days
SELECT
  id,
  name,
  expiry_date,
  stock,
  supplier_id
FROM medicines
WHERE expiry_date BETWEEN CURDATE() AND DATE_ADD(CURDATE(), INTERVAL 30 DAY)
ORDER BY expiry_date ASC;

-- Query 4: Calculate monthly revenue
SELECT
  DATE_FORMAT(created_at, '%Y-%m') AS month,
  COUNT(*) AS order_count,
  SUM(total_amount) AS total_revenue,
  AVG(total_amount) AS avg_order_value
FROM orders
WHERE status = 'delivered'
GROUP BY DATE_FORMAT(created_at, '%Y-%m')
ORDER BY month DESC;

-- Query 5: Get pending prescriptions with patient details
SELECT
  p.id,
  u.name,
  u.email,
  u.phone,
  p.uploaded_at,
  p.file_original_name
FROM prescriptions p
JOIN users u ON p.user_id = u.id
WHERE p.status = 'pending'
ORDER BY p.uploaded_at ASC;

-- Query 6: Find drug interactions for selected medicines
SELECT DISTINCT
  i.id,
  m1.name AS medicine1,
  m2.name AS medicine2,
  i.severity,
  i.description
FROM interactions i
JOIN medicines m1 ON i.medicine1_id = m1.id
JOIN medicines m2 ON i.medicine2_id = m2.id
WHERE i.medicine1_id IN (?) OR i.medicine2_id IN (?)
ORDER BY i.severity DESC;
```

7.5 Performance Optimization

7.5.1 Query Optimization

```
Optimization Techniques:
1. Index Usage:
   - Use indexes on frequently searched columns
   - Maintain index statistics
   - Regular index maintenance

2. Query Optimization:
   - Use EXPLAIN to analyze queries
   - Avoid N+1 queries with JOINs
   - Use pagination for large result sets
   - Cache frequently accessed data

3. Denormalization:
   - Cache price_at_purchase in ORDER_ITEMS (for reporting)
   - Cache user_email in ORDERS (for notifications)
   - Cache medicine_name in ORDER_ITEMS (for receipts)

4. Database Tuning:
   - Increase buffer pool size
   - Optimize query cache
   - Tune connection pool size
   - Monitor slow query log
```

7.5.2 Scalability Considerations

```
Horizontal Scaling:
- Database replication for read scaling
- Write operations on primary, reads on replicas
- Sharding strategy for very large datasets

Vertical Scaling:
- Increase server resources (CPU, RAM)
- Better hardware for faster I/O
- SSD storage for faster queries

Archival Strategy:
- Archive old orders (> 2 years) to separate storage
- Keep 2 years active data in primary database
- Faster queries on smaller active dataset
```

APPENDIX A: Technology Stack

A.1 Backend Technologies

- **Runtime**: Node.js 18+
- **Framework**: Express 4.19.2
- **Database**: MySQL 8.0
- **Authentication**: JWT (jsonwebtoken)
- **Password Hashing**: bcryptjs
- **Payment**: Stripe API
- **Email**: Nodemailer (SMTP)
- **Rate Limiting**: express-rate-limit
- **Security**: Helmet
- **Validation**: express-validator

A.2 Frontend Technologies

- **Library**: React 18.2.0
- **Router**: React Router 6.8.0
- **Forms**: React Hook Form
- **HTTP Client**: Axios
- **State Management**: Context API
- **Notifications**: React Toastify
- **Styling**: CSS Modules
- **Build Tool**: Create React App

A.3 DevOps & Deployment

- **Version Control**: Git/GitHub
- **CI/CD**: GitHub Actions
- **Hosting**: AWS/DigitalOcean
- **Containerization**: Docker
- **Monitoring**: ELK Stack / Datadog
- **APM**: New Relic

APPENDIX B: Glossary

Term	Definition
API	Application Programming Interface for system communication
JWT	JSON Web Token for stateless authentication
ERD	Entity Relationship Diagram showing database structure
CRUD	Create, Read, Update, Delete operations

SLA	Service Level Agreement for system availability
RTO	Recovery Time Objective for disaster recovery
RPO	Recovery Point Objective for data loss tolerance
GDPR	General Data Protection Regulation for data protection
COD	Cash on Delivery payment method
SMTP	Simple Mail Transfer Protocol for email sending
3NF	Third Normal Form for database normalization
UUID	Universally Unique Identifier

APPENDIX C: Change Log

Version	Date	Changes
1.0	Nov 2025	Initial document creation

CONCLUSION

This comprehensive project documentation provides a complete blueprint for the Online Medical Store Management System development. The document includes:

1. **Introduction**: Purpose, benefits, and stakeholder context
2. **Scope**: Clear boundaries of what's included and excluded
3. **Requirements**: Detailed functional and non-functional specifications
4. **Use Cases**: Complete actor and system interaction model
5. **Work Plan**: Timeline with Gantt chart and milestones
6. **ERD**: Complete database relationships
7. **Database Design**: Detailed schema with optimization strategies

This documentation serves as the foundation for development, testing, and maintenance phases of the project.

Document Prepared By: Development Team
Date: November 2025
Version: 1.0
Status: Complete

End of Document